Highly Efficient Phosphine-Free $Pd(OAc)_2$-Catalyzed Fukuyama Coupling Reaction: Synthesis of a Key Intermediate for (+)-Biotin under Low Catalyst Loading

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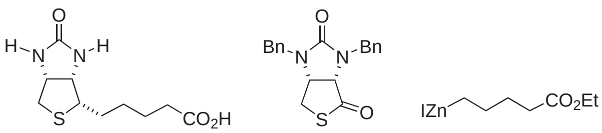
Abstract: A phosphine-free catalytic system involving inexpensive $Pd(OAc)_2$ in conjunction with Zn dust and $ZnBr_2$ was unexpectedly found to effect the Fukuyama coupling reaction of thiolactone **2** with zinc reagent **3** to provide vinyl sulfide **6**, a key intermediate for (+)-biotin (**1**), in high yield with a very tiny amount of Pd (up to 0.01 mol%, TON = 6800).

Key words: zinc, palladium, catalysis, cross-coupling, (+)-biotin

The Pd-catalyzed cross coupling reaction has recently aroused considerable attention as a synthetic method due to the wide spread use for assembling a variety of unique and imposing architecture of valuable compounds, such as drugs and natural products.[1] We have previously developed an efficient synthesis of (+)-biotin (**1**) through installation of 4-carboxybutyl chain by the Fukuyama coupling reaction[2] of thiolactone **2** with zinc reagent **3** (Figure 1).[3]



1 2 3

Figure 1

Screening of the required metal catalyst has been extensively undertaken and heterogeneous Pd/C has proven to be most effective, owing to ease of recovery and low cost. A mechanistic advantage of the reaction employing Pd/C as the catalyst is to help Pd highly disperse on the solid support to allow the formation of active monomeric or oligomeric Pd species without agglomerating to inactive Pd black. A similar efficient reaction may thus be developed as well in the homogeneous system by designing a catalytic system that allows the formation of the active monomeric or oligomeric Pd species during the reaction. The idea has built momentum that has led to our unexpected result, in that the coupling reaction of **2** with **3** did work through the phosphine-free Fukuyama coupling reaction employing an extremely small amount of readily available $Pd(OAc)_2$ (up to 0.01 mol%).

In the Pd/C-catalyzed coupling reaction of thiolactone **2** with zinc reagent **3**, a considerable reduction of the required iodide **4** has been accomplished by the addition of Zn dust and $ZnBr_2$, generated in situ from Zn dust and Br_2 (Table 1, entries 1–3).[3f] The reaction employing a homogeneous Pd catalyst without adding any phosphine ligand was thus tested in the presence of Zn dust and $ZnBr_2$. Zn dust (2.7 equiv) was first activated by the addition of Br_2 (0.7 equiv), which concurrently generates $ZnBr_2$. Addition of iodide **4** to the mixture provided a suspension involving zinc reagent **3**, activated Zn dust and $ZnBr_2$. Then, $Pd(OAc)_2$ (1.2 mol%), thiolactone **2** and DMF were subsequently added, and the mixture was stirred at 30–35 °C for 20 h. Gratifyingly, the reaction did take place and treatment of the mixture with aqueous HCl afforded the desired vinyl sulfide **6**[3a] in 98% yield (TON = 82, Table 1, entry 4). The use of $PdCl_2$ in place of $Pd(OAc)_2$ resulted in a poor yield (11% yield, Table 1, entry 5). Noteworthy are the observations that either addition of PPh_3 (4.8 mol%) to the mixture or employment of phosphine-including Pd catalyst, $PdCl_2(PPh_3)_2$[2a] was detrimental to the reaction, providing **6** in 9% and 36% yield, respectively (Table 1, entries 6 and 7). Reduction of the catalyst loading was investigated, and, in the presence of 0.01 mol% of $Pd(OAc)_2$, compound **6** was obtained in 68% yield with TON of 6800 (Table 1, entry 9). To the best of our knowledge, the TON obtained is the highest value so far recorded in the Pd-catalyzed synthesis of ketones from carboxylic acid derivatives.[4–6]

Although the mechanism of the present reaction remains to be seen, the use of Zn dust and $ZnBr_2$ is significant for the reaction. Monomeric or oligomeric catalytically active species may be effectively formed from $Pd(OAc)_2$ by the addition of the additives under low catalytic loading. The use of a very small amount of Pd catalyst is advantageous as well for not aggregating to inactive black Pd.

In conclusion, a new protocol for the synthesis of a key intermediate for (+)-biotin was worked out, which involves homogeneous Pd catalyst system involving $Pd(OAc)_2$, Zn and $ZnBr_2$. Elimination of phosphine ligand as well as the

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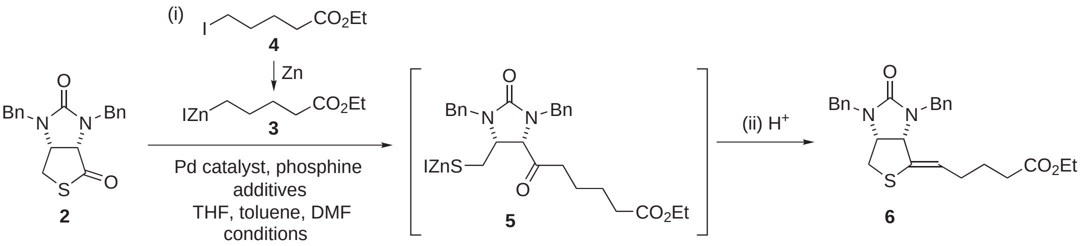
Table 1 The Fukuyama Coupling Reaction of Thiolactone **2** with Ethoxycarbonylbutylzinc Iodide (**3**) in the Presence of Various Pd Catalysts and Additives



Entry	Pd catalyst (mol%)[a]	Phosphine (mol%)[a]	Additives (equiv)[b]	Iodide **4** (equiv)[b]	Conditions	Conversion[c] (%)	Assay yield[c] (%)	TON
1[3f]	Pd/C[d] (0.86)	–	Zn (3.6), TMSCl (0.25)	2.5	25–30 °C, 18 h	–[e]	95	110
2[3f]	Pd/C[d] (0.86)	–	Zn (2.7), TMSCl (0.14)	1.4	25–30 °C, 18 h	–[e]	59	69
3[3f]	Pd/C[d] (0.86)	–	Zn (2.7), Br_2 (0.7)	1.4	25–30 °C, 18 h	–[e]	94	109
4	$Pd(OAc)_2$ (1.2)	–	Zn (2.7), Br_2 (0.7)	1.4	30–35 °C, 20 h	98	98	82
5	$PdCl_2$ (1.2)	–	Zn (2.7), Br_2 (0.7)	1.4	30–35 °C, 20 h	13	11	9
6	$Pd(OAc)_2$ (1.2)	PPh_3 (4.8)	Zn (2.7), Br_2 (0.7)	1.4	30–35 °C, 20 h	10	9	8
7	$PdCl_2(PPh_3)_2$ (1.2)	–	Zn (2.7), Br_2 (0.7)	1.4	30–35 °C, 20 h	41	36	30
8	$Pd(OAc)_2$ (0.1)	–	Zn (2.7), Br_2 (0.7)	1.4	30–35 °C, 20 h	93	90	900
9	$Pd(OAc)_2$ (0.01)	–	Zn (2.7), Br_2 (0.7)	1.4	30–35 °C, 20 h	75	68	6800

[a] Mol% relative to thiolactone **2**.
[b] Equivalent relative to thiolactone **2**.
[c] Determined by HPLC.
[d] Pd/C catalyst was purchased from Degussa Japan Co., Ltd. and has the following properties: impregnation depth, 50–150 nm (egg shell); reduction degree, 0–25%; Pd dispersion, 29%; water content, ca. 3 wt%.
[e] Not determined.

quite low catalyst loading would permit an alternative and practical access to (+)-biotin. Elucidation of the reaction mechanism and further application to the synthesis of multifunctional ketones[7] are under investigation, which will be reported elsewhere in due course.

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(6) **Synthesis of 6 from 2 (Table 1, Entry 8).**
To a suspension of zinc powder (3.57 g, 54.6 mmol) in a mixed solvent of THF (7.0 mL) and toluene (4.7 mL) was added bromine (0.72 mL, 14.0 mmol) below 40 °C for 5 min. After the mixture was stirred at 25 °C for 30 min, it was warmed up to 50 °C, and ethyl 5-iodopentanoate (**4**, 7.16 g, 28.0 mmol) was added at 50–55 °C for 30 min. The mixture was stirred at the same temperature for 1 h and cooled down to 30 °C. To the mixture were added thiolactone **2** (6.77 g, 20.0 mmol), toluene (9.3 mL), $Pd(OAc)_2$ (4.5 mg, 0.02 mmol) and DMF (1.7 mL, 22.0 mmol), and the mixture was stirred at 30–35 °C for 20 h. The mixture was cooled down

to 5 °C, and 3 N HCl (21 mL) was added to the mixture below 20 °C. The mixture was stirred at 25 °C for 30 min and it was filtered. To the filtrate was added EtOAc (30 mL), and the organic layer was warmed up to 45 °C and evaporated. A part of the residue was analyzed by HPLC [L-Column, MeCN–THF–0.03 M phosphate buffer (K_2HPO_4/KH_2PO_4 = 1:1) = 40:8:52, 40 °C, 1 mL/min, 254 nm]. The assay yield of **6** was determined to be 90%. The residue was purified by silica gel column chromatography (toluene–EtOAc = 15:1) to give **6** as slight yellow oil (7.87 g, 87%). The spectroscopic data of the product were in complete accordance with those of the reported value.[3a]

(7) A preliminary investigation has shown that the present protocol is applicable to the synthesis of multifunctional ketones from other thiol esters than **2**.